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book reviews

Plasticity

Dynamic Plasticity. By N. Cristescu. John Wiley & Sons, Inc., New York, N. Y., 1968. xi and 614 pages. $25.

REVIEWED BY R. J. CLIFTON[1]

THIS BOOK is a rewritten and expanded edition of a book published in Romanian in 1958. The overall objective of this book is the presentation of solutions and methods of solution for boundary value problems in dynamic plasticity. To this end, a wide variety of problems are considered, for a fairly wide variety of types of assumed material behavior. The problem areas considered include: longitudinal plastic wave propagation (rate independent and rate dependent), the mechanics of extensible strings, the mechanics of circular membranes, axisymmetric problems, plane waves, shock waves in solids, and plastic waves in soils. The appendix contains a presentation of various differential constitutive equations which are used in dynamic plasticity. There is also an extensive bibliography.

The solutions are generally obtained by the method of characteristics or by the numerical solution of difference equations obtained from incremental relations along characteristics. The numerical solutions would be of greater interest if experimental data were available for the conditions that the theoretical models are intended to simulate; however, these solutions do serve to illustrate the essential features of the theoretical predictions. The book should be of considerable interest to researchers and engineers interested in the dynamic plastic deformation of metals.

Thermodynamics

Thermodynamics. By R. Kubo in cooperation with H. Ichimura, T. Usui, and N. Hashitsume. John Wiley & Sons, Inc., New York, N. Y., 1968. xii and 300 pages. $16.

REVIEWED BY J. KESTIN[2]

THE appearance of this book constitutes a rare, though by no means unique, event in that it presents us with an English translation (by the authors themselves) of a major scientific work from the Japanese. In the original, the present booklet appeared as the first part of a more comprehensive book on Thermodynamics and Statistical Mechanics. The section on Statistical Mechanics appeared in English earlier under the title *Statistical Mechanics—An Advanced Course* (North Holland, 1965).

Although the content of the book is strictly classical, both from the point of view of the selection of subject matter and method of presentation, it constitutes a useful addition to the already large literature on the subject, owing to its clarity and unusual conciseness. Judged in American terms, the booklet can be described as one in Classical Thermodynamics or Physical Chemistry. Naturally, aiming at being brief, the authors could not include many applications, and those relevant to engineering do not go beyond the study of Carnot cycles. Much of what can be described as applications is treated in a large selection of worked and unworked examples which grace the book.

An engaging feature of the book is a series of twelve *Divertissements* in which the authors, so to say, relax, and make "off-the cuff" comments on thermodynamics. These range from a vignette on the founders of the First Law of Thermodynamics to musings on thermodynamics and life. The authors also include a passage on "why do we have winter heating" modeled on a similar one in your reviewer's translation of A. Sommerfeld's *Thermodynamics and Statistical Mechanics* (Academic Press, 1956). Though thoughtful, the argument is unfortunately erroneous, or at best, incomplete, because it neglects to take into account the fact that the air in a leaky, heated room constitutes an open rather than a closed thermodynamic system.

It is useful to have this booklet on one's shelf, but its high price will act as a deterrent against wide circulation.

Lubrication

Standard Handbook of Lubrication Engineering. By J. J. O'Connor. McGraw-Hill Book Company, Inc., New York, N. Y., 1968. xxii and 1030 pages. $27.50.

REVIEWED BY E. A. SAIBEL[3]

THIS HANDBOOK contains a great deal of information much of which has been scattered throughout the literature and hence has been difficult to find. It is divided into four main sections: 1, Lubrication Principles; 2, General Lubrication Engineering Practices; 3, Lubrication of Specific Equipment; and 4, Lubrication in Specific Industries. The last one covers the steel industry, nonferrous industries, railroads, nuclear power plants, and certain space problems. The list of chapter headings is long and impressive. It is sufficient here to note that almost all aspects of lubrication—fluid, gas, solid, and boundary—are covered.

As might be expected from its size, some 1000 pages, and from the number of authors, about 60, there are variations in quality and in extent of coverage, but in the main it is clear and well done. If additional information is required, references are copious; but in the majority of cases the engineer will find all he needs for the design or analysis of conventional sliding or rolling elements in this book.

This work is one to which every engineer concerned with lubrication of any sort should have access.

Acoustics

Theoretical Acoustics. By Philip M. Morse and K. Uno Ingard. McGraw-Hill Book Company, New York, N. Y., 1969. xix and 927 pages. $23.

REVIEWED BY R. T. BEYER[4]

IN 1936, Professor Morse published his highly successful *Vibration and Sound*, which gave a concise mathematical formulation of the basic problems of vibrating strings, bars, and plates, as well as of the radiation and scattering of sound waves in air. This volume, together with the second edition which appeared in 1948, served as a fundamental work of instruction and reference for a full generation.

The current text is a greatly expanded version of the earlier editions. In comparison with its predecessors, this volume

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compresses the material on bars, membranes, and plates, more than doubles the space devoted to radiation and scattering, and adds a number of new topics. The space devoted to the different topics is in line with their significance in modern acoustics. Thus the chapters on sound in ducts and rooms and on the coupling of acoustical systems reflect the importance of these problems in architectural acoustics and noise control; the chapters on the acoustics of moving media and on nonlinear acoustics treat problems of importance in modern fluid dynamics.

The text is strongly mathematical and should appeal to those who have found the earlier editions useful. The chapters on plasma acoustics and Brillouin scattering treat topics of importance in physical acoustics, but the analysis here too is from the formal mathematical viewpoint, rather than from basic physical considerations.

The book contains a considerable number of relevant problems to be worked by the reader. It is to be regretted that it does not also possess references to recent journal literature in this field.

Shock and Vibration

Vibration and Shock in Damped Mechanical Systems. By J. C. Snowdon. John Wiley & Sons, Inc., New York, N. Y., 1968. xiv and 486 pages. $18.95.

REVIEWED BY D. I. G. JONES[5]

FEW recent books have been devoted exclusively to consideration of the response of damped structures to shock and vibration. Professor Snowdon's book not only contains a wealth of detailed information concerning the damping properties of rubberlike materials and vibration isolation, mechanical impedance, and wave propagation in damped beams and rods, but it also serves as a valuable introductory text for those concerned with analysis of the effect of damping on the response of more complex structures. The book is very well written and illustrated.

Viscoelasticity

Viscoelasticity. By Wilhelm Flügge. Blaisdell Publishing Company, Waltham, Mass., Toronto, London, 1967. viii and 127 pages.

REVIEWED BY J. BESSELING[6]

THE AUTHOR has presented us with an admirably concise and consistent introductory text on linear viscoelasticity. It is aimed at a reader who is familiar with the basic concepts of mechanics, including stress and strain in two dimensions, and the technique of deriving differential equations from the consideration of the mechanics of an infinitesimal element. On the mathematical side, the prerequisite is calculus and a brief exposure to complex numbers and to linear differential equations. Laplace transformation, integral equations, partial differential equations, and complex contour integration are explained as far as needed.

The various chapters of the book contain an exceptionally clear exposition on viscoelastic models, hereditary integrals, viscoelastic beams, vibrations, axial impact, buckling of columns, and viscoelasticity in three dimensions. For illustrative examples results are derived which are of interest for engineering tasks, but the author has omitted purposely reference to specific applications, which, in the author's words, are here today and gone tomorrow, to make room for others. Thereby the whole space of this little book could be devoted to a discussion of the basic viscoelastic phenomena and of the various tools of mathematical analysis. The author succeeds in combining generality with utmost simplicity, for instance, in introducing the concept of complex compilance and in discussing the correspondence principle.

The book can be strongly recommended to teachers of a course on this special subject of viscoelasticity, but it will be found particularly useful by those who want to widen their field of knowledge in mechanics, but who so often are being discouraged by unbalanced or learned treatises on the subject. For self-study, the exercises at the end of each chapter are well chosen.

Optimization

Optimization in Control Theory and Practice. By I. Gumowski and C. Mira. Cambridge University Press, 1968. viii and 242 pages. $10.50.

REVIEWED BY A. E. PEARSON[7]

THIS book is a useful addition to the growing library of books on optimization. Particular emphasis is given to the treatment of extremal problems by the calculus of variations, although one chapter is devoted to the relationships between the calculus of variations and the techniques of dynamic programming and the maximum principle. Chapter 1, which is entitled "The Gap Between Control Theory and Practice," provides an interesting discussion on the problem of modeling a dynamical system and its importance in formulating extremal problems. Chapters 2 and 3 contain a résumé of the known results of the calculus of variations with special reference to the work of Carathéodory.

A noteworthy feature of this presentation is the generous display of examples which the authors use to emphasize the importance of the property of semicontinuity of functional to be minimized, the effect of the domain of definition of a functional on the existence of solutions, and a number of topics relating to the existance and uniqueness of solutions to extremal problems. Unfortunately, there is little attempt to reconcile the gap between theory and practice as it was rather systematically discussed in the first chapter. In addition, the formulation of extremal problems is carried out in the context of the classical calculus of variations rather than in the context of optimal control theory, and none of the examples in the book are drawn from the field of control systems. Therefore, the book will be mainly of interest to the research engineer and applied mathematician seeking an appendage of results in the classical calculus of variations.

Structures

Theory of Inelastic Structures. By T. H. Lin. John Wiley & Sons, Inc., New York, N. Y., 1968. xiv and 454 pages. $19.95.

REVIEWED BY T. H. H. PIAN[8]

THE present book is dealing with the analysis of structures with inelastic strains which, according to the author's definition, includes all non-Hookian strains such as thermal, time-independent plastic and time-dependent creep strains. The methods developed in this book are based on an extended Duhamel's analogy by treating a structure involving thermal strain and/or other inelastic strains as an identical elastic structure with an additional set of equivalent body forces.

The book consists of 12 chapters. Chapter 1 presents a review of stress and strain and governing equations for an elastic solid and Chapter 2 introduces the analogy between inelastic strains,

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